

Name:Supriya.B Register Number: 250701760 Experiment Name: Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# -----
# 1. CREATE DATASET
# -----
data = {
    "Date": [
        "01-01-2023", "02-01-2023", "03-01-2023", "04-01-2023", "05-01-2023",
        "06-01-2023", "07-01-2023", "08-01-2023", "09-01-2023", "10-01-2023",
        "11-01-2023", "12-01-2023", "13-01-2023", "14-01-2023", "15-01-2023",
        "16-01-2023"
    ],
    "Product": [
        "Product A", "Product B", "Product A", "Product C", "Product B", "Product A",
        "Product C", "Product B", "Product A", "Product C", "Product B", "Product A",
        "Product C", "Product B", "Product A", "Product C"
    ],
    "Sales": [
        200, 150, 220, 300, 180, 210, 320, 160,
        230, 310, 190, 240, 330, 170, 250, 340
    ],
    "Quantity": [
        4, 3, 5, 6, 4, 5, 7, 3,
        6, 7, 4, 6, 8, 3, 7, 8
    ],
    "Region": [
        "North", "South", "North", "East",
        "West", "North", "East", "South",
        "North", "East", "West", "North",
        "East", "South", "North", "East"
    ]
}
# Create DataFrame
df = pd.DataFrame(data)
# Convert Date column
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)
# -----
# 2. DISPLAY DATA
# -----
print("DATASET:")
print(df.head())
print("\n")
# -----
# 3. CHECK MISSING VALUES
```

```
# ----- print("MISSING VALUES:")
print(df.isnull().sum())
print("\n")
```

```

# ----- # 4. STATISTICAL
SUMMARY
# ----- print("STATISTICAL
SUMMARY:")
print(df[["Sales", "Quantity"]].describe()) print("\n")
# ----- # 5. PRODUCT-WISE SALES
AND QUANTITY
# ----- product_summary =
df.groupby("Product")[["Sales", "Quantity"]].sum().reset_index()
print("PRODUCT-WISE SALES AND QUANTITY:")
print(product_summary)
print("\n")
# ----- # 6. LINE GRAPH - SALES
OVER DATE
# ----- plt.figure(figsize=(10, 5))
plt.plot(
    df["Date"],
    df["Sales"],
    marker="o",
    color="blue",
    linewidth=2
)
plt.title("Sales Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
# ----- # 7. BAR GRAPH -
PRODUCT-WISE SALES
# ----- plt.figure(figsize=(8, 5))
plt.bar(
    product_summary["Product"],
    product_summary["Sales"],
    color=["blue", "orange", "green"]
)
plt.title("Product-wise Total Sales")
plt.xlabel("Product")
plt.ylabel("Total Sales")
plt.tight_layout()
plt.show()
# ----- # 8. CORRELATION MATRIX
# ----- correlation = df[["Sales",
"Quantity"]].corr() print("CORRELATION MATRIX:")
print(correlation)
print("\n")
# ----- # 9. CORRELATION
HEATMAP
# ----- plt.figure(figsize=(7, 5))
sns.heatmap(
    correlation,
    annot=True,

```

```
cmap="coolwarm",  
linewidths=0.5,  
fmt=".2f"  
)  
plt.title("Correlation Matrix")  
plt.tight_layout()  
plt.show()
```

DATASET:

```
Date Product Sales Quantity Region 0 2023-01-01 Product A  
200 4 North 1 2023-01-02 Product B 150 3 South 2 2023-01-03  
Product A 220 5 North 3 2023-01-04 Product C 300 6 East 4  
2023-01-05 Product B 180 4 West
```

MISSING VALUES:

```
Date 0  
Product 0  
Sales 0  
Quantity 0  
Region 0  
dtype: int64
```

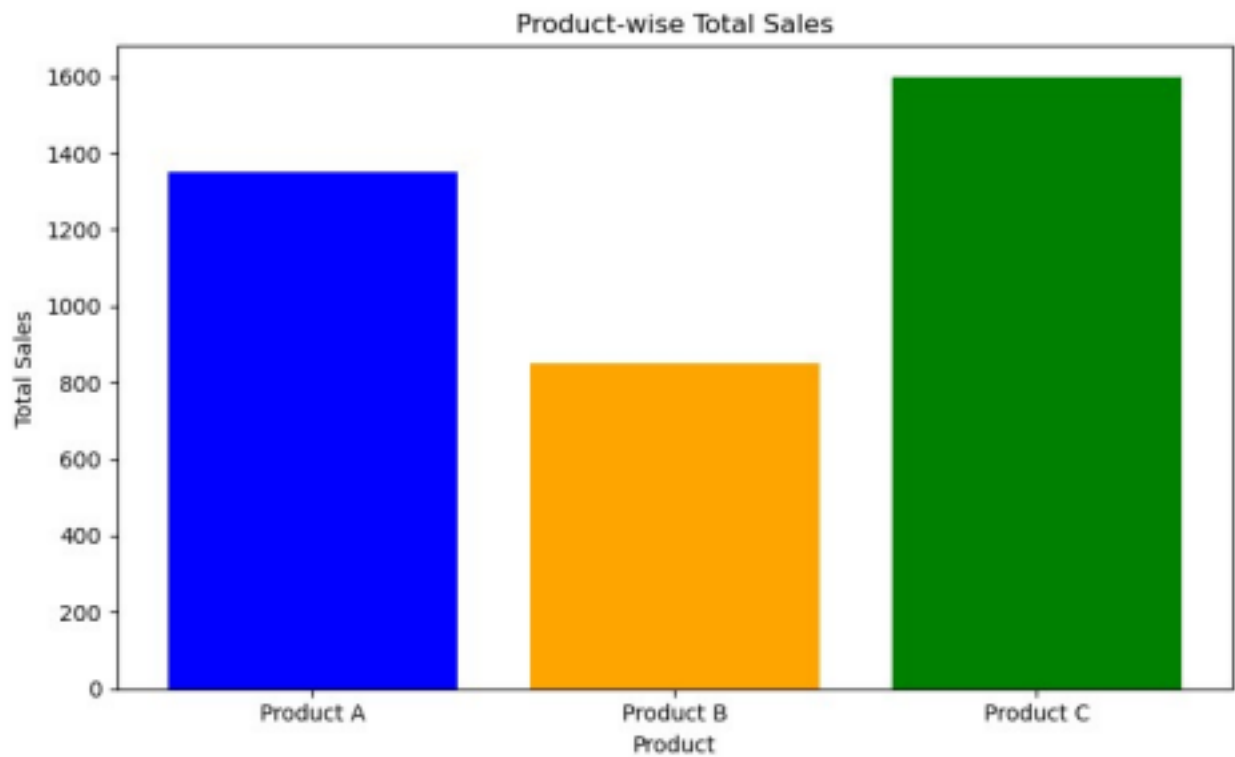
STATISTICAL SUMMARY:

```
Sales Quantity  
count 16.000000 16.000000  
mean 237.500000 5.375000  
std 64.031242 1.746425  
min 150.000000 3.000000  
25% 187.500000 4.000000  
50% 225.000000 5.500000  
75% 302.500000 7.000000
```

```
max 340.000000 8.000000
```

PRODUCT-WISE SALES AND QUANTITY:

```
Product Sales Quantity  
0 Product A 1350 33  
1 Product B 850 17  
2 Product C 1600 36
```



CORRELATION MATRIX:
Sales Quantity
Sales 1.000000 0.944922
Quantity 0.944922 1.000000

